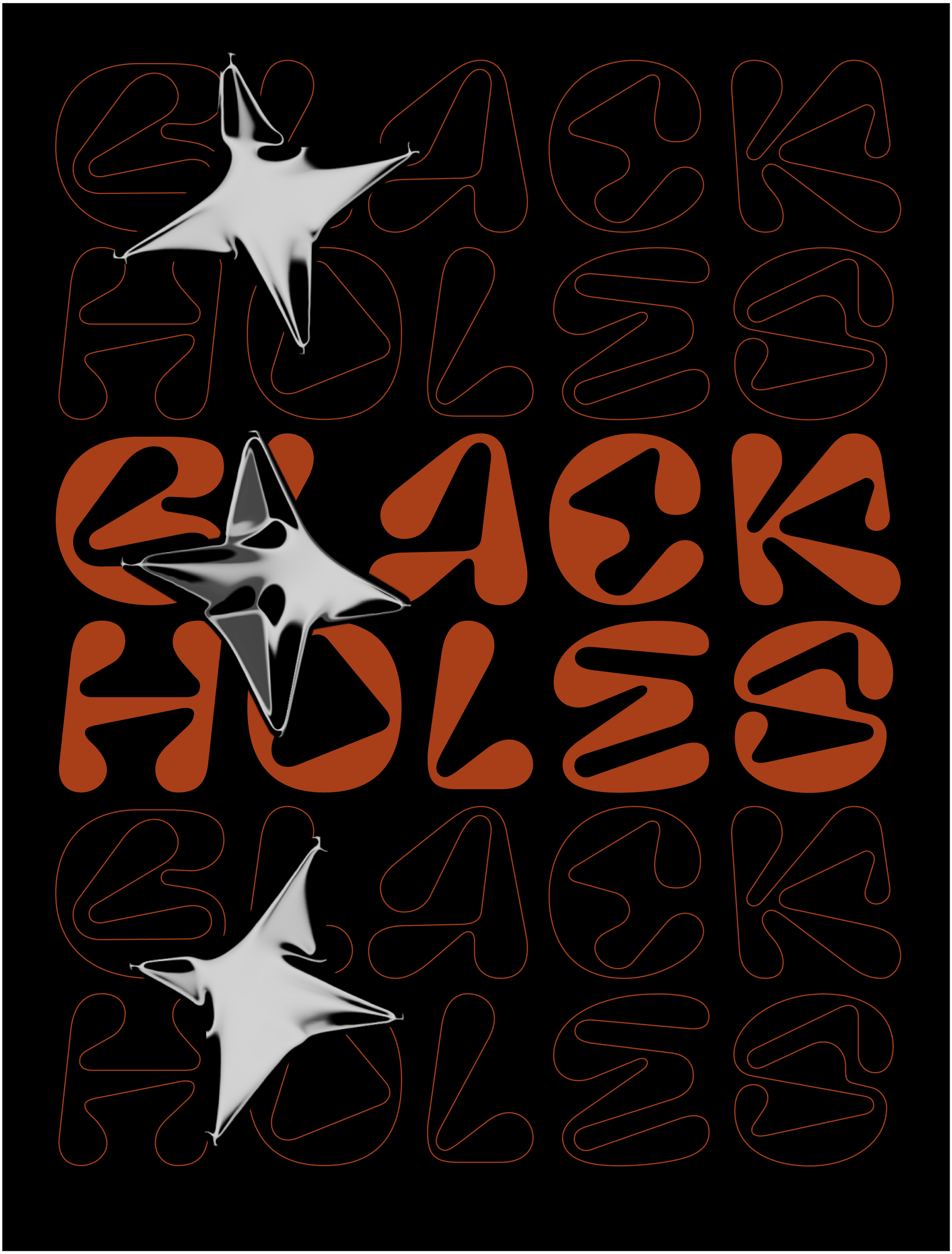
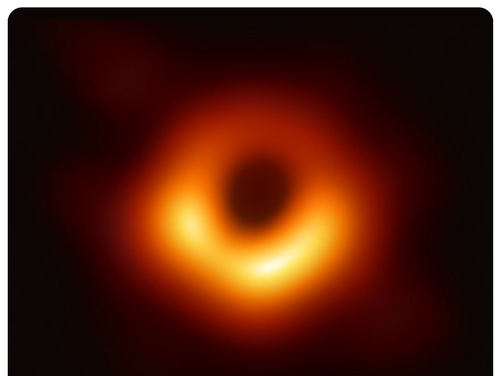

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Black holes are regions in space where an enormous amount of mass is packed into a tiny volume. This creates a gravitational pull so strong that not even light can escape. They are created when giant stars collapse, and perhaps by other methods that are still unknown.

Black holes fascinate both the public and scientists—they push the limits of our understanding about matter, space and time.

Scientists at the University of Chicago and across the world have made many discoveries about our universe with the help of black holes, but there's a lot we still don't know about these extraordinary cosmic phenomena.



What is a black hole?

Black holes are made of matter packed so tightly that gravity overwhelms all other forces.

When you pick up a bowling ball, it's heavy because the matter is densely packed. If you packed more and more mass into the same tiny space, eventually it would create gravity so strong that it would exert a significant pull on passing rays of light.

Black holes are created when massive stars collapse at the end of their lives (and perhaps under other circumstances that we don't know about yet). One of the first steps toward the discovery of black holes was made by University of Chicago professor Subrahmanyan Chandrasekhar, when he realized that massive stars would have to collapse after they ran out of fuel for the fusion reactions which keep them hot and bright.

The universe is full of black holes. In the past decade, scientists have detected the signals of their collisions and taken images of the light from the gas swirling around them—and this has helped us learn many things about the universe. For example, black holes have helped us test Einstein's theory of general relativity, which describes how mass, space, and time are related to one another. Scientists think they can tell us much more about these and other essential rules of the universe.



And on a more personal level, the supermassive black hole at the center of our own Milky Way galaxy may have played a role in how Earth came to be here!

What do black holes look like?

Black holes themselves are invisible—they emit virtually no light and so cannot be seen directly. But we have developed several ways to find them anyway.

By looking for the stuff that's falling in. If material is falling into a black hole, it travels at such high speeds that it gets hot and glows very brightly, and we can detect that. (That's how the Event Horizon Telescope took its famous first images of black holes.) Scientists hope to use this method to learn a lot more about how and what black holes "eat."

By seeing their gravity pulling on other things. We can find black holes by watching the movements of visible objects around them. For example, a black hole's gravity is so strong that nearby stars will orbit around them, so we can look for stars behaving strangely around a patch of "empty" space. From this, we can calculate exactly how heavy that black hole must be. That's how Nobel Prize winner Andrea Ghez and her team detected the supermassive black hole at the center of our own galaxy.

By detecting the gravitational ripples when they collide. We can also detect black holes by detecting the ripples in space-time created when two of them crash into each other. From that signal, we can tell how massive the black holes were, how far away they were, and how fast they were traveling when they collided.

What's inside a black hole?

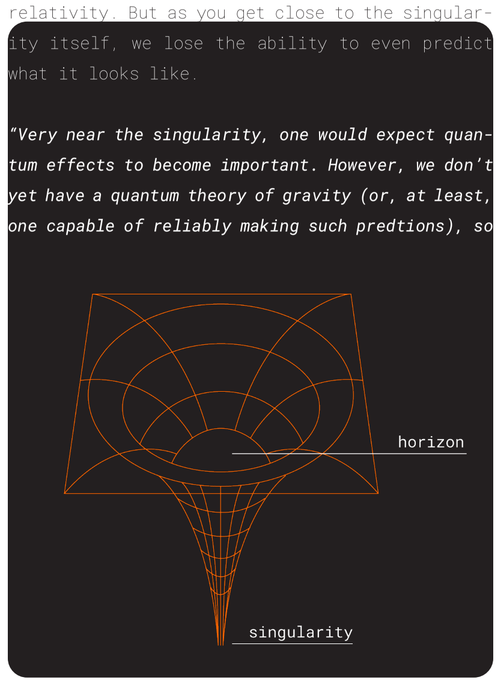
The short answer is that no one knows!

"In some ways that's one of the most profound questions in physics," said University of Chicago Prof. Daniel Holz. "There are not many cases in physics where we simply cannot predict what happens, but this is one of them."

Black holes have two parts. There is the event horizon, which you can think of as the surface, though it's simply the point where the gravity gets too strong for anything to escape. And then, at the center, is the singularity. That's the word we use to describe a point that is infinitely small and infinitely dense.

We have a good understanding of what the event horizon looks like, thanks to the laws of general relativity. But as you get close to the singularity itself, we lose the ability to even predict what it looks like.

"Very near the singularity, one would expect quantum effects to become important. However, we don't yet have a quantum theory of gravity (or, at least, one capable of reliably making such predtions), so we just don't know the correct description of the singularity—or even whether it really is a singularity," said University of Chicago Prof. Robert Wald.



Scientists think that black holes eventually will explode, but it will take many, many times longer than the current age of the universe for that to happen. What will it look like when that happens? That's another big mystery.

"Maybe there's a little nugget left behind containing all of the information that fell into the black hole, maybe there's a portal to a new universe, maybe the information is just gone forever; we simply don't know," said Holz.

(If all of this is unsatisfying, know that it keeps scientists awake at night, too.)

How do black holes form?



Scientists know about one way that black holes form, but there may be others.

One way to make a black hole is to have a massive star collapse at the end of its life. Prof. Subrahmanyan Chandrasekhar was the first to calculate that when a massive star burns up all its fuel, it will collapse. The idea was ridiculed at first, but other scientists calculated that the star continues forever to fall inward toward its center—thus creating what we called a black hole.

Black holes can grow more massive over time as they "eat" gas, stars, planets and even other black holes!

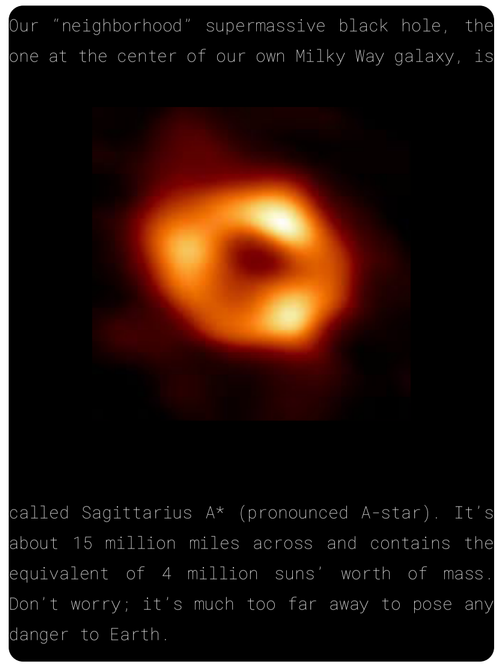
There's another type of black hole called a supermassive black hole. These are way too massive to have been created by one star collapsing; it's still a mystery how they form. Black holes can eat other black holes, so it's possible that the supermassive ones are made of many small black holes merged together. *"Or perhaps these big black holes were especially hungry, and ate so much of their surroundings that they grew to enormous size,"* said Prof. Holz. But we can see these supermassive black holes formed very early on in the universe—maybe too early to have been made by stars getting old enough to collapse—so it's possible there's some other way to make a black hole that we don't know about yet.

What is a supermassive black hole?

There are two kinds of black holes: star-sized black holes and supermassive black holes.

Supermassive black holes are so named because they contain on the order of millions to billions times the mass of our sun.

As far as we can tell, nearly every galaxy in the universe has one of these supermassive black holes sitting right at its center like a seed. And they are correlated—a bigger galaxy has a bigger black hole, and a smaller galaxy has a smaller black hole. All of this makes scientists think these supermassive black holes have something to do with how the galaxies formed. But that relationship is still a mystery, and so is how the supermassive black holes formed in the first place.



Our "neighborhood" supermassive black hole, the one at the center of our own Milky Way galaxy, is called Sagittarius A* (pronounced A-star). It's about 15 million miles across and contains the equivalent of 4 million suns' worth of mass. Don't worry; it's much too far away to pose any danger to Earth.

What do black holes eat?

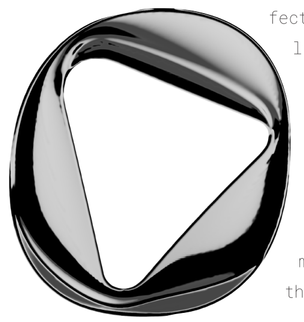
Contrary to what you may have seen in movies, black holes don't actually "suck" things in. For example, there are actually stars orbiting the supermassive black hole at the center of our galaxy, and they'll keep orbiting without falling in unless something else disturbs them. An object really has to fall right into the mouth of a black hole for it to be eaten. (And the mouth, which we call the event horizon, of a black hole, is tiny; if the entire Earth were to collapse and form a black hole, its mouth would be less than an inch across!)

But the movements of stars and galaxies do sometimes mean that stuff falls into a black hole's mouth. Sagittarius A*, the black hole at the center of our galaxy, mostly eats interstellar gas and dust that is drifting around. With telescopes, we have seen other black holes eating stars and even the gas from neighboring galaxies.

Black holes can be "messy eaters." As objects are being ripped apart, some of the gas and matter can be flung off at high speeds. Sometimes this is so powerful that it forms jets and winds shooting outwards at nearly the speed of light, and this can affect the galaxy containing it. These jets can blow apart nearby stars and planets; or they can provide just the right amount of churn to create the ideal conditions for making new stars over millions of years.

What do black holes tell us about the universe?

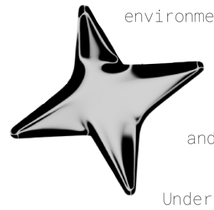
Black holes are kind of like a playground for physicists. *"They are literally made out of space and time,"* said Prof. Holz. Because they are so extreme, they are the perfect place to test the limits of the rules of the universe.



Observing them and thinking about their properties have yielded enormous insights about the nature of the universe. For example, detecting their collisions allowed us to test Einstein's theories about how mass, space, and time are related (as well as lots of other theories about the universe). Black holes also seem to play a role in the formation of galaxies; it's likely our supermassive black hole has something to do with how we came to be here today.

Some other things we can learn about the universe include:

Understanding extreme physics and how stars and planets grow. Some supermassive black holes are extremely active, gobbling up stars amid swirling magnetic fields and flinging out jets of superheated gas and material; these systems are known as quasars. Watching this process can tell us about the physics of these extreme environments. It can also tell us about the conditions under which stars, planets, and galaxies are born, grow, and die.



Understanding how fast the universe is expanding and therefore how it evolved. As we get more and more data on pairs of black holes colliding, Holz and others have worked out methods to use them to calculate how fast the universe is expanding. This number, called the Hubble Constant, is key to understanding the past, present, and future behavior of the universe, as well as the nature of dark matter and dark energy.



Reconciling our major theories of the universe. One of the most fundamental questions in modern physics is how to reconcile quantum mechanics, which is the law for the very smallest particles in the universe, with general relativity, which is the law for the very biggest things in the universe. These two sets of laws don't quite match up. But black holes are the perfect place to explore the links between them.



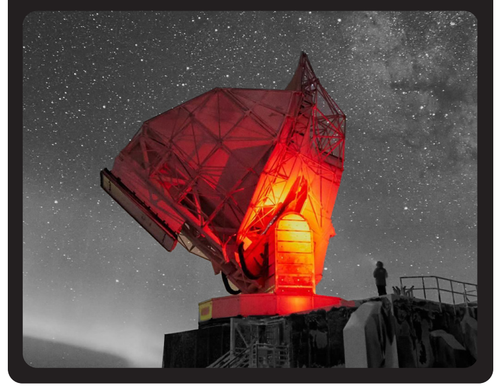
For example, Stephen Hawking theorized that the laws of quantum mechanics suggest that black holes have a very tiny temperature—which was surprising to scientists, since that implies some radiation is leaving the black hole. This has all sorts of implications for our understanding of physics. (One such implication: Black holes should eventually lose mass faster and faster over time until they explode. However, that will take trillions upon trillions of years to happen, so none of us will be around when it does.)

How was the first picture of a black hole taken?

In 2019, people around the world were thrilled to see the first image ever taken of a black hole—and then, in 2022, an image of our "personal" supermassive black hole in the Milky Way. The bright ring around each one is created by material glowing very hot as it circles the black hole.

This was the first direct image of a black hole ever taken—all of the other pictures you've seen are simulations or artist illustrations.

These black holes are so far away that no normal telescope would ever be powerful enough to see them. You would need a telescope the size of the Earth—but scientists figured out that they could piece together images taken simultaneously from telescopes situated all around the Earth instead. (One of these was the South Pole Telescope, run by a collaboration headed by the University of Chicago, which provided the view from the bottom of the Earth.)



All together, this network of combined telescopes is referred to as the Event Horizon Telescope. They next hope to create a "movie" of the glowing gas moving around a black hole as it's pulled in.

What do scientists still not know about black holes?

Even as new detectors and telescopes have been able to tell us more and more about black holes in the past decades, scientists still have hundreds of questions about black holes. What do they eat, and how often? What happens as stuff falls in? When it falls in, how much comes back out? Does this stuff end up causing the black hole to spin? How are these black holes created in the first place?



There are more fundamental questions, too, ranging from 'What's inside a black hole?' to 'How are supermassive black holes tied to their galaxies?'

"Everything about black holes is absurd. It's very appealing to say they can't possibly exist, except that both our theories and our observations show that they must and in fact do exist," said Prof. Holz.

One thing that keeps scientists awake at night is whether information that falls into a black hole is truly gone forever. There are other laws of physics that say that all information in the universe is preserved; even if you burn a notebook, its information could theoretically be recovered from the traces and gases that are left behind, as well as the light that was emitted. But as far as we can tell, it's possible that the information within a notebook dropped into a black hole could be truly erased from the universe.

Mheslinga. (2025, June 13). Black holes, explained. University of Chicago News. https://news.uchicago.edu/explainer/black-holes-explained